

YINUO HAN

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RESEARCH INTERESTS

- Exoplanetary systems, debris disks, planet formation, disk substructure, dynamical evolution
- High-angular-resolution imaging, high-contrast optical/infrared imaging, millimeter interferometry
- Dust formation, carbon feedback, Wolf-Rayet stars, colliding-wind binaries

ACADEMIC APPOINTMENTS

1. **Melza M. and Frank Theodore Barr Fellow, Caltech, USA** Since Oct 2024
Postdoctoral fellowship, Planetary Sciences, Division of Geological and Planetary Sciences

EDUCATION

1. **PhD in Astronomy, University of Cambridge, UK** Oct 2020 - Jul 2024
Institute of Astronomy · Trinity College · Gates Cambridge Scholar
Thesis: *The Structure of Extrasolar Planetesimal Belts in Images*
Advisor: Prof. Mark Wyatt
2. **BSc (Adv) (Hons), University of Sydney, Australia** Feb 2016 - Dec 2019
Majors in Physics, Neuroscience · First Class Honours (Physics) and the University Medal
Thesis: *A Dance with Dragons: The Enigmatic Wolf-Rayet Binary Apep*
Advisor: Prof. Peter Tuthill
- Study Abroad, Physics, University of California, Berkeley, USA** Aug 2017 - Dec 2017

ADVISING

List of theses and research projects advised.

- *VLT observations of β Pic and its dust clump.* Jun – Sep 2025
Summer research project/Caltech SURF by Polaris Hayes (Caltech).
- *ALMA imaging of WR112 and dust modeling with JWST/MIRI.* Jun – Sep 2025
Summer research project/Caltech SURF by Donglin Wu (Yale University).
- *Joint ALMA and JWST/NIRCam modeling of the vertical structure of the AU Mic debris disk.* Jun – Sep 2025
Summer research project/Caltech SURF by Junyi Zhao (University of Michigan).
- *Secular perturbation as the origin of the dust and gas clump of Beta Pictoris.* Oct 2022 – Jun 2023
Cambridge Part III/Masters project by Faran Sarwar (co-advised with Mark Wyatt).
- *The structure of the Beta Pictoris debris disk inferred from multi-wavelength mid-infrared images.* Jul – Aug 2022
Summer research project by Andrew Zhang.

TEACHING

1. **Topics in Astrophysics** (Distributions, Timescales, Tides, Planet Formation) 2023
Institute of Astronomy, University of Cambridge
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
2. **Structure and Evolution of Stars** 2022
Institute of Astronomy, University of Cambridge
Regular supervisions/tutorial classes for Part II Astrophysics/final-year undergraduate astrophysics.
3. **Oscillating Systems, Waves and Quantum Waves, Fields** 2021
Trinity College, University of Cambridge
Weekly supervisions/tutorial classes for Part IA Physics/first-year undergraduate physics.
4. **Physics 1B** (Electromagnetism, Fluids, Quantum Mechanics) 2019
School of Physics, University of Sydney
Weekly tutorial classes for first-year undergraduate physics.

OBSERVING

- *Keck P.I. program*: Probing planet–disk interactions in asymmetric debris disks 2026B
- *ALMA co-I. program*: Testing a new picture of debris disk gas by observing CI towards HD 121617 (ID 2025.1.00273.S, P.I. Gianni Cataldi) C12, 2025
- *VLT/VISIR P.I. program*: Testing the giant impact scenario to explain the dust clump and cat’s tail in Beta Pictoris (114.27CQ) P114, 2024
- *ALMA P.I. program*: First resolved ALMA imaging of carbon-rich Wolf-Rayet binary dust structures (2024.1.00803.S) C11, 2024
- *ALMA co-I. program*: Resolving the extended structure associated with Apep’s enigmatic wind (2024.1.01244.S, P.I. Joseph Callingham) C10, 2024
- *JWST P.I. program*: What lies beyond the inner spiral of Apep? (GO 5842) C3, 2024
- *JWST P.I. program*: What causes warm dust interior to planetesimal belts? (GO 5709) C3, 2024
- *VLT/VISIR P.I. program*: Wind kinematics in the extreme colliding-wind binary Apep: the first Galactic long-duration gamma-ray burst progenitor? (113.26A7) P113, 2024
- *ALMA P.I. program*: First ALMA imaging of dust around Wolf-Rayet binaries (2023.1.00999.S) C10, 2023
- *ALMA co-I. program*: Vertical structure and planetary system dynamics (2023.1.00487.S and 2025.1.00362.S, P.I. Meredith Hughes) C10/C12, 2023/2025
- *JWST co-I. program*: Fingerprinting the history of episodic dust creation in Wolf-Rayet binaries (4093, P.I. Noel Richardson) C2, 2023
- *Australia Telescope Compact Array*: Executing observations of Wolf-Rayet binary Apep 2019

PRESS

- *NRAO press release based on Donglin Wu's summer research published in Wu et al. 2026.* Feb 2026
Yale press release.
- *Caltech press release based on Han et al. 2026.* Jan 2026
Other ARKS collaboration press releases from NRAO, ALMA and ERC.
- *NASA, STScI and Caltech press releases based on Han et al. 2025.* Nov 2025
Featured on Astronomy Picture of the Day.
- *University of Cambridge press release based on Han et al. 2022.* Oct 2022
Reported by major science and mainstream news outlets (e.g., ABC, BBC, CNN, The Guardian).
- *University of Sydney press release based on Han et al. 2020.* Oct 2020
Reported by 110 news items from 21 countries in 8 languages (e.g., CNN).

ACADEMIC AWARDS

UNIVERSITY OF CAMBRIDGE

- Gates Cambridge Scholarship* 2020
- Cambridge International Scholarship*, accepted Gates above instead 2020

UNIVERSITY OF SYDNEY

- University Medal*, top award among graduating class in physics 2019
- Henry Chamberlain Russell Prize for Astronomy*, for best astronomy Honours thesis 2019
- University of Sydney Honours Scholarship* 2019
- Faculty of Science Olympiad Scholarship* 2016 - 2019
- Deas-Thomson Scholarship in Physics*, for top of class in third-year physics 2018
- W.I.B. Smith Prize in Experimental Physics* 2018
- Dean's List of Academic Excellence* 2016, 2017, 2018
- Sydney Scholars Award* 2016 - 2018
- Denison Summer Research Scholarship* 2016, 2017
- Sydney Abroad International Exchange Scholarship*, supported second-year physics exchange 2017
- University of Sydney Academic Merit Prize* 2016
- Levey Scholarship in Physics*, for top of class in first-year physics 2016
- Smith Prize in Experimental Physics* 2016

OTHER

- University of Oxford Clarendon Scholarship*, declined 2020
- International Physics Olympiad Bronze Medal* 2015

RESEARCH GRANTS

EXTERNAL GRANTS

- P.I., JWST Program 5842 (USD 78k) 2024 – 2027
- P.I., JWST Program 5709 (USD 74k) 2024 – 2027

INTERNAL GRANTS

Barr Fellowship research grant, Caltech (USD 30k)	2024 – 2027
Gates Cambridge Scholarship research funding, University of Cambridge (approx. USD 3k)	2020 – 2024
Rouse Ball research funding, Trinity College, Cambridge (approx. USD 4k)	2023

TALKS

CONFERENCE TALKS

- Discs on the Exe Jul 2026
Probing debris disk vertical dynamics with multi-wavelength imaging. Exeter, UK
- Debris Disk Connections Jul 2026
A JWST view of a radially broad collisional cascade in the debris disk of γ Ophiuchi. Cambridge, UK
- Exoplanets and Planet Formation Dec 2025
The high-resolution structure of debris disks in the ARKS ALMA program. Shanghai, China
- Exploring Star–Planet–Disk Connections Oct 2025
High-resolution ALMA imaging of extrasolar planetesimal belts. Tartu, Estonia
- Europlanet Science Congress Sep 2024
The high-resolution structure of debris disks revealed by the ARKS ALMA program. Berlin, Germany
- Raising the veil on star formation near and far Apr 2024
High-resolution ALMA imaging of debris disks with the ARKS program. Cambridge, UK
- Dust Devils: Debris Disks in the Sonoran Desert Mar 2024
The radial structure of debris disks in the ARKS ALMA program. Tucson, USA
- Cosmic Dust Aug 2023
The physics of dusty Wolf-Rayet binaries revealed by observations of WR140. Kitakyushu, Japan
- UK Exoplanet Community Meeting Sep 2022
Recovering the structure of edge-on debris disks non-parametrically. Edinburgh, UK
- Debris Disks at Home and Abroad Sep 2022
Recovering the structure of edge-on debris disks non-parametrically. Jena, Germany

SEMINARS

- Macquarie University May
Observing extrasolar comets Sydney, Australia
- Institute for Astronomy, University of Hawaii Apr 2026
Understanding planetary system formation and evolution through the lens of debris disks Manoa HI, USA
- Caltech DIX Planetary Science Seminar Mar 2026
Imaging Extrasolar Kuiper Belts and Interpreting Their Structures Pasadena CA, USA
- MPA Planet and Star Formation Seminar Oct 2025
The formation and evolution of dust in the colliding-wind binary Apep revealed by JWST. Heidelberg, Germany
- Tsinghua Exoplanets and Planet Formation Seminar Sep 2025
Imaging debris disks with ALMA and JWST. Beijing, China

- Columbia Astronomy & Astrophysics Seminar
Resolving radial substructures in debris disks. Mar 2025
New York City, USA
- JPL Astrophysics Luncheon Seminar
Interpreting the Resolved Structure of Debris Disks. Feb 2025
Pasadena, USA
- Department of Applied Mathematics and Theoretical Physics, University of Cambridge
The radial and vertical structure of debris disks. Oct 2023
Cambridge, UK
- Institute of Astronomy, University of Cambridge
Stellar fireworks: the spiral dust plumes of Wolf-Rayet binaries. Oct 2022
Cambridge, UK
- Institute of Astronomy, University of Cambridge
A non-parametric method to recover the structure of edge-on debris disks. Mar 2022
Cambridge, UK

OUTREACH TALKS

- Institute of Astronomy Open Evening
What are exoplanets doing? Asteroid belts give us a clue. Oct 2023
Cambridge, UK
- Institute of Astronomy Open Evening
Stellar fireworks: a journey through the spirally dust of massive binary stars. Jan 2023
Cambridge, UK
- Integrated Science Class Seminar (Year 6–8), Ming Cheng School
Planets of the Solar System and beyond. Apr 2022
Beijing, China
- Gates Cambridge Teach-a-thon
Planets of the Solar System and beyond. Feb 2021
Online
- Australian Council of Undergraduate Research
Highly anisotropic stellar winds in recently-identified Wolf-Rayet binary. Oct 2019
Newcastle, Australia

POSTER PRESENTATIONS

- Gordon Research Conference on the Origins of Solar Systems
The radial structure of debris disks in the ARKS ALMA program. Jun 2025
South Hadley, USA
- Gordon Research Seminar + Conference on the Origins of Solar Systems
Modeling the structure of debris disks and the dust clump of Beta Pictoris. Jun 2023
South Hadley, USA
- STScI Spring Symposium: Planetary Systems and the Origins of Life in the Era of JWST
Modeling the structure of debris disks and the dust clump of Beta Pictoris. May 2023
Baltimore, USA
- Protostars and Planets VII
Has the dust clump in the debris disk of Beta Pictoris moved? Apr 2023
Kyoto, Japan

PUBLICATIONS

Publications available on ORCID 0000-0002-2106-0403, ADS Library and Google Scholar.

FIRST-AUTHOR PUBLICATIONS

9. **Y. Han**, M. C. Wyatt, K. Y. L. Su, *et al.* A radially broad collisional cascade in the debris disk of γ Oph revealed by JWST. *The Astrophysical Journal* (2026). doi.org/10.3847/1538-4357/ae6ef7

8. **Y. Han**, M. C. Wyatt, M. Jankovic, *et al.* The multiwavelength vertical structure of the archetypal β Pictoris debris disk. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202659709
7. **Y. Han**, E. Mansell, J. Jennings, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) II. The Radial Structure of Debris Disks. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556450
6. **Y. Han**, R. M. T. White, J. R. Callingham, *et al.* The formation and evolution of dust in the colliding-wind Wolf-Rayet binary Apep observed by JWST. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/ae12e5
5. **Y. Han**, M. C. Wyatt, S. Marino. Recovering the structure of debris disks non-parametrically from images. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf282
4. **Y. Han**, M. C. Wyatt, W. R. F. Dent. Has the dust clump in the debris disk of Beta Pictoris moved? *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stac3769
3. **Y. Han**, P. G. Tuthill, R. M. Lau, A. Soulain. Radiation-driven acceleration in the expanding WR140 dust shell. *Nature* (2022). doi.org/10.1038/s41586-022-05155-5
2. **Y. Han**, M. C. Wyatt, L. Matr . RAVE: a non-parametric method for recovering the surface brightness and height profiles of edge-on debris disks. *Monthly Notices of the Royal Astronomical Society* (2022). doi.org/10.1093/mnras/stac373
1. **Y. Han**, P. G. Tuthill, A. Soulain, J. R. Callingham, P. M. Williams, P. A. Crowther, B. J. S. Pope, B. Marcote. The extreme colliding-wind system Apep: resolved imagery of the central binary and dust plume in the infrared. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa2349

CO-AUTHORED PUBLICATIONS

* Indicates projects led by students who I advised

26. [In press] E. Chiang, T. D. Pearce, M. R. Jankovic, A. J. Backues, **Y. Han**, A. V. Krivov, M. Pan, B. Zawadzki, *et al.* Viscously Stirring Particle Disks into Lorentzians and Gaussians to Infer Dynamical and Collisional Masses (ARKS XIII). *The Astrophysical Journal* (2026). arxiv.org/abs/2604.19743
25. J. Olofsson, M. R. Jankovic, S. Marino, A. V. Krivov, M. Bonduelle, G. Cataldi, **Y. Han**, A. M. Hughes, T. L hne, S. Mac Manamon, E. Mansell, L. Matr , J. Milli, A. A. Sefilian, P. Th bault, B. Zawadzki, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) XI: Gas-dust interactions and radial offsets between micron and millimetre-sized grains. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556489
24. * D. Wu, **Y. Han**, *et al.* Constraining properties of dust formed in Wolf-Rayet binary WR 112 using mid-infrared and millimeter observations. *The Astrophysical Journal* (2026). doi.org/10.3847/1538-4357/ae31f1
23. S. Marino, L. Matr , A. M. Hughes, J. Ehrhardt, G. M. Kennedy, C. del Burgo, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) I. Motivation, Sample,

- Data Reduction and Results Overview. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556489
22. B. Zawadzki, A. Fehr, A. M. Hughes, E. Mansell, J. Kittling, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) III. The Vertical Structure of Debris Discs. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556505
 21. S. Mac Manamon, L. Matrà, S. Marino, A. Brennan, **Y. Han**, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) IV. CO gas imaging and overview. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556605
 20. J. Milli, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) V. Comparison between scattered light and thermal emission. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556523
 19. J. B. Lovell, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VI. Asymmetries and Offsets. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556568
 18. A. Brennan, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VII. Optically Thick Gas with Broad CO Gaussian Local Line Profiles in the HD 121617 Disc. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556566
 17. S. Marino, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) VIII. A dust arc and non-Keplerian gas kinematics in HD 121617. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556493
 16. P. Weber, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) IX. Gas-driven origin for the continuum arc in the debris disc of HD121617. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556855
 15. M. Jankovic, *et al.* The ALMA survey to Resolve exoKuiper belt Substructures (ARKS) X. Interpreting the peculiar dust rings around HD 131835. *Astronomy & Astrophysics* (2026). doi.org/10.1051/0004-6361/202556637
 14. R. M. T. White, B. J. S. Pope, P. G. Tuthill, **Y. Han**, *et al.* The Serpent Eating Its Own Tail: Dust Destruction in the Apep Colliding-Wind Nebula. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-4357/adfbel
 13. J. D. Monnier, **Y. Han**, *et al.* Revealing the Accelerating Wind in the Inner Region of Colliding-wind Binary WR 112. *The Astrophysical Journal* (2025). doi.org/10.3847/1538-3881/adfa03
 12. N. D. Richardson, *et al.* Carbon-rich dust injected into the interstellar medium by Galactic WC binaries survives for hundreds of years. *The Astrophysical Journal* (2025). 10.3847/1538-4357/addf30
 11. M. Sommer, M. C. Wyatt, **Y. Han**. A PR drag origin for the Fomalhaut disk’s pervasive inner dust: constraints on collisional strengths, icy composition, and embedded planets. *Monthly Notices of the Royal Astronomical Society* (2025). doi.org/10.1093/mnras/staf494

10. E. P. Lieb, *et al.* Dynamic imprints of colliding-wind dust formation from WR140. *The Astrophysical Journal Letters* (2025). doi.org/10.3847/2041-8213/ad9aa9
9. R. M. Lau, *et al.* A first look with JWST aperture masking interferometry (AMI): resolving circumstellar dust around the Wolf-Rayet binary WR 137 beyond the Rayleigh limit. *The Astrophysical Journal* (2024). doi.org/10.3847/2041-8213/ad9aa9
8. J. Terrill, S. Marino, R. A. Booth, **Y. Han**, J. Jennings, M. C. Wyatt. Deprojecting and constraining the vertical thickness of exoKuiper belts. *Monthly Notices of the Royal Astronomical Society* (2023). doi.org/10.1093/mnras/stad1847
7. R. M. Lau, *et al.* From dust to nanodust: resolving circumstellar dust from the colliding-wind binary Wolf-Rayet 140. *The Astrophysical Journal* (2022). doi.org/10.3847/1538-4357/acd4c5
6. R. M. Lau, M. J. Hankins, **Y. Han**, *et al.* Nested dust shells around the Wolf-Rayet binary WR 140 observed with JWST. *Nature Astronomy* (2022). doi.org/10.1038/s41550-022-01812-x
5. P. A. Robinson, X. Gao, **Y. Han**. Relationships between lognormal distributions of neural properties, activity, criticality, and connectivity. *Biological Cybernetics* (2021). doi.org/10.1007/s00422-021-00871-z
4. B. Marcote, J. R. Callingham, M. De Becker, P. G. Edwards, **Y. Han**, R. Schulz, J. Stevens, P. G. Tuthill. AU-scale radio imaging of the wind collision region in the brightest and most luminous non-thermal colliding wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2021). doi.org/10.1093/mnras/staa3863
3. R. M. Lau, M. J. Hankins, **Y. Han** *et al.* Resolving periodic spirals and shadows from the Wolf-Rayet dust factory WR112. *The Astrophysical Journal* (2020). doi.org/10.3847/1538-4357/abaab8
2. J. R. Callingham, P. A. Crowther, P. M. Williams, P. G. Tuthill, **Y. Han**, B. J. S. Pope. Two Wolf-Rayet stars at the heart of colliding-wind binary Apep. *Monthly Notices of the Royal Astronomical Society* (2020). doi.org/10.1093/mnras/staa1244
1. M. Li, **Y. Han**, M. J. Aburn, M. Breakspear, R. Q. Poldrack, J. M. Shine, J. T. Lizier. Transitions in information processing dynamics at the whole-brain network level are driven by alterations in neural gain. *PLOS Computational Biology* (2019). doi.org/10.1371/journal.pcbi.1006957